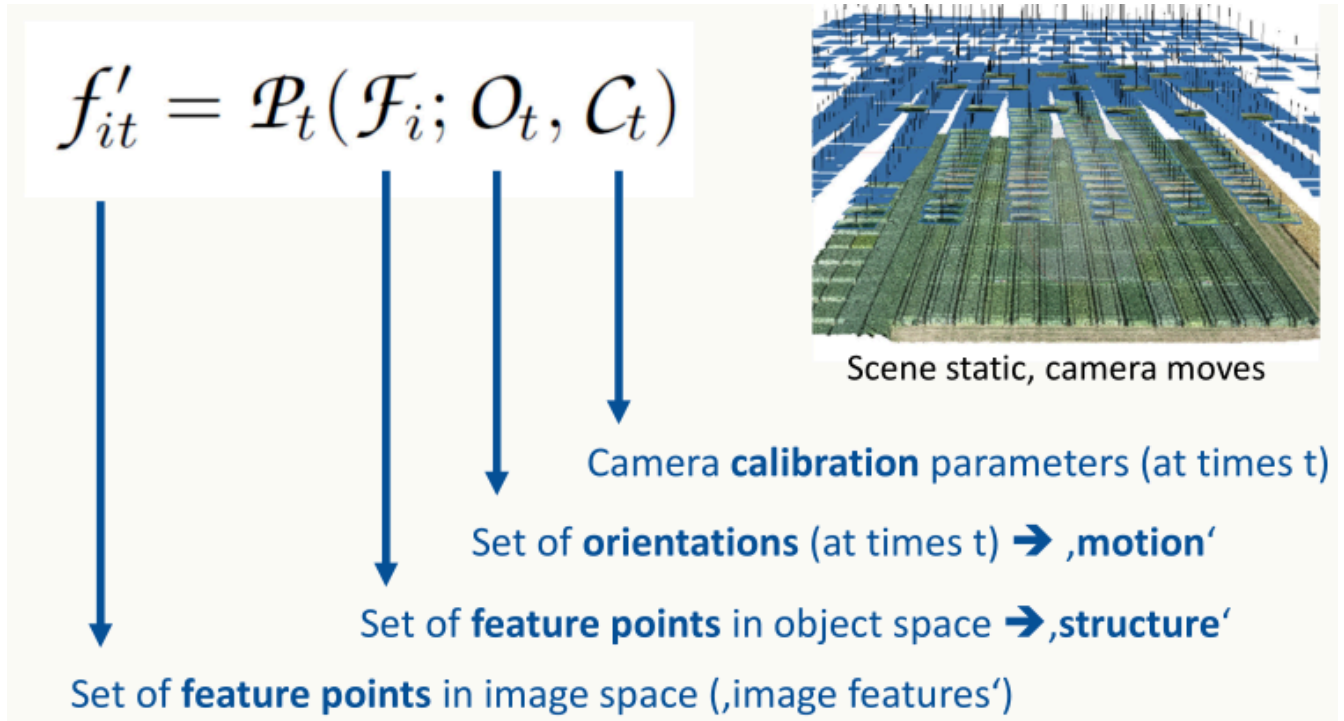


L7: SfM - Images to Point Clouds

- Before Mobile Laser Scanning
- For 3D Positions: Images from single observation not enough
- Definition from Förstner's Book



- Orientation or **motion from structure**
 - get $\{O_i\}$ from correspondences $\{F_i, f_{it}\}$
- **Calibration**
 - get C_t from $\{f_{it}\}$ and possibly given $\{F_i\}$
- Reconstruction or **structure from motion (SfM)**
 - get $\{F_i\}$ from $\{f_{it}\}$
- **Relative orientation**
 - get relative pose between multiple images and **local** scene description $\{F_i\}$ from corresponding image points $\{f_{i_a} f_{i_b} f_{i_c} \dots\}$ in multiple images a, b, c, ...
- **Absolute Orientation:**
 - Get **global** scene $\{F_i\}$ from **local** scene $\{F_i\}$ and 3D control features

Bundle Adjustment - Förstner's Book

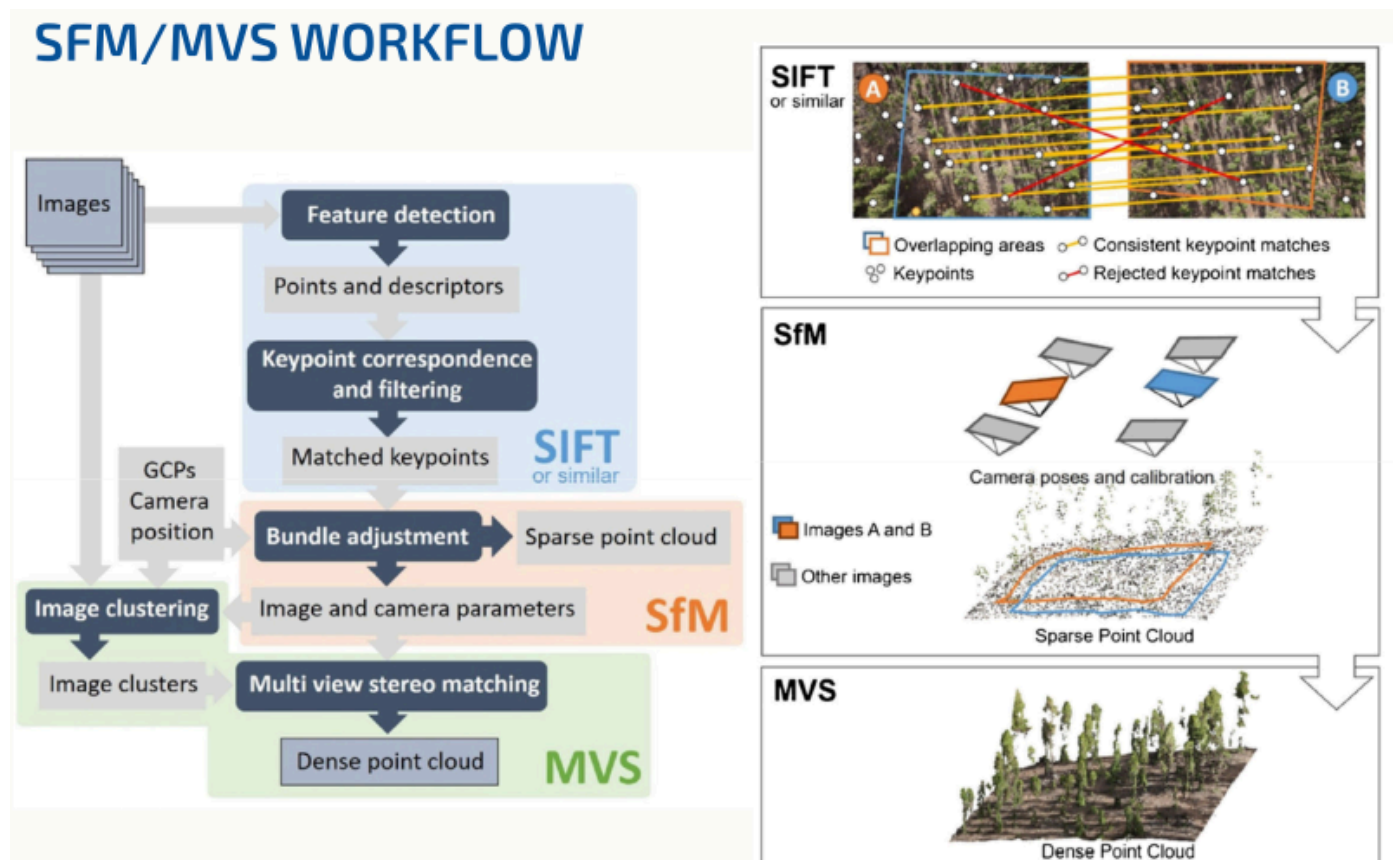
- get $\{F_i\}$ and $\{O_i\}$ and C_T from $\{f_{it}\}$
- **global** $\{F_i\}$ also needs 3D control features

- or direct measurements of some poses/points (e.g. via GNSS)
- Closely related to SLAM

SfM actually:

- BA no necessarily scaled and in absolute coordinates
- Result: Sparse Point cloud $\{F_i\}$ and image parameters $\{O_i\}, C_t$

SfM/MVS workflow



• SIFT

1. Extract Features
 - SIFT, SURF, BRISC, ...
 - Corner or Blob Features
 - Also Multi-scale approaches
2. Feature description
 - Vector of numbers describing gradients, intensities, ...
3. Feature matching
 - Finding same Features in multiple images
 - Outlier rejection

• SfM

1. Initialise

- Use 2 images
- Initialise structure

2. Add new images

- Orient new image using existing structure
- Run local BA
- Initialise new structure points

3. Optimal Georeferencing

1. Include information from GPS or Ground control points

4. Run final BA on complete image block

• MVS - Multi View Stereo

- Try to find much more correspondences between images than using features
- Ideal case: match every pixel
- Match **areas** instead of features
- Use **known orientations** and epipolar constraints
- Use multi-resolution
- Use multi-image matching
- Use assumptions on surface smoothness
 - semi-global matching
- -> Result: **Dense Cloud**

Georeferencing

- Georeferencing is needed
 - Get scale OR
 - get scene in a global coordinate system

• Options:

1. Option: Indirect Georeferencing

- classical case
- Use known scene points to **absolutely orient all images** and scene
- Deploy Ground Control points and measure coordinates
 - Use total stations: ~mm accuracy
 - RTK GNSS: ~1-3cm accuracy

2. Option: Direct Georeferencing

- Rarely used

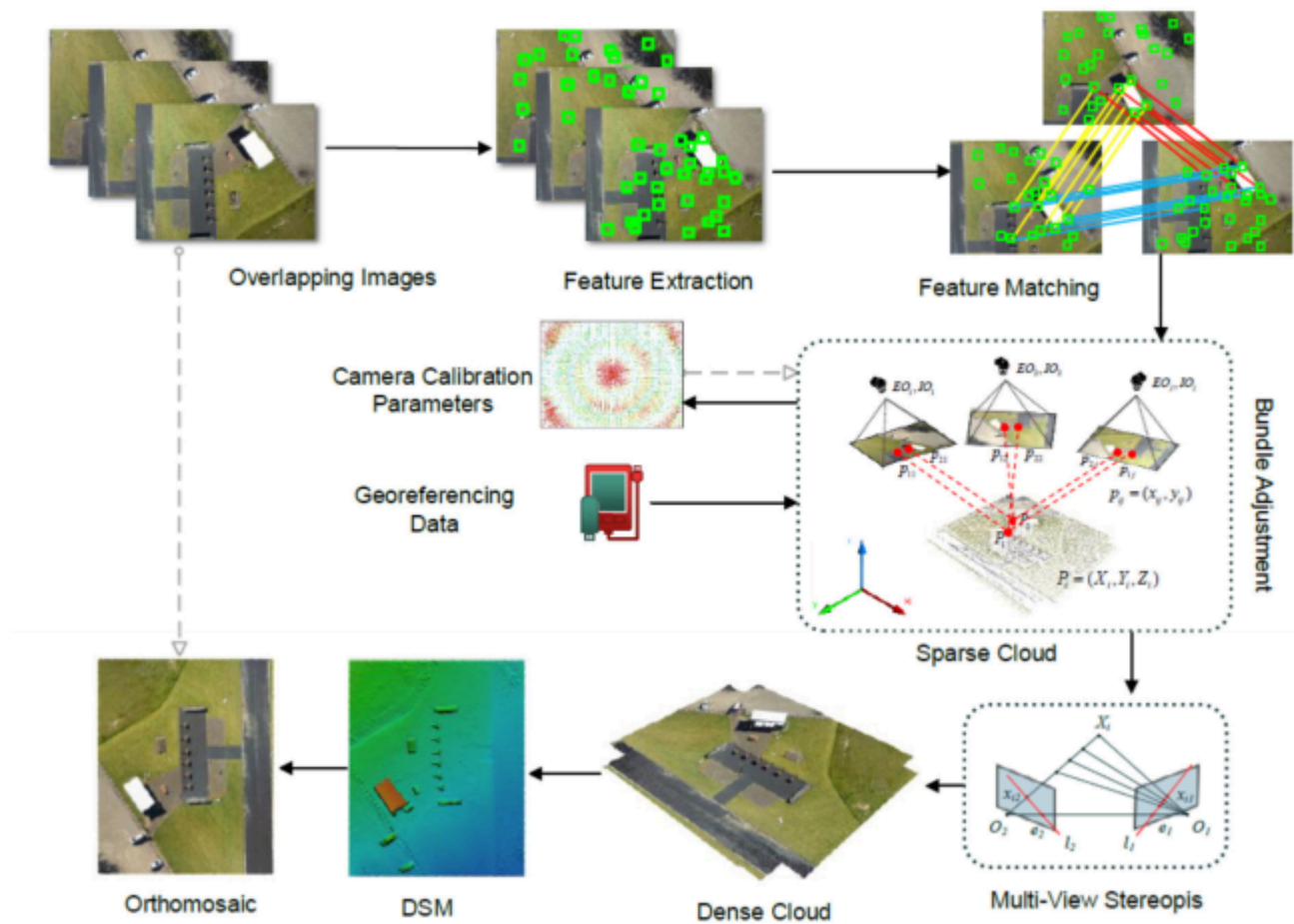
- Usage of onboard sensors for absolute orient images and skip SfM step
- Determine **camera pose** at measurement
- Required:
 - System/algorithm to determine pose of UAV: **Trajectory estimation**
 - Relative pose between UAV and camera: **System Calibration**
 - Synchronisation between: **Sensor Synchronisation**

3. Option: Integrated Sensor Orientation

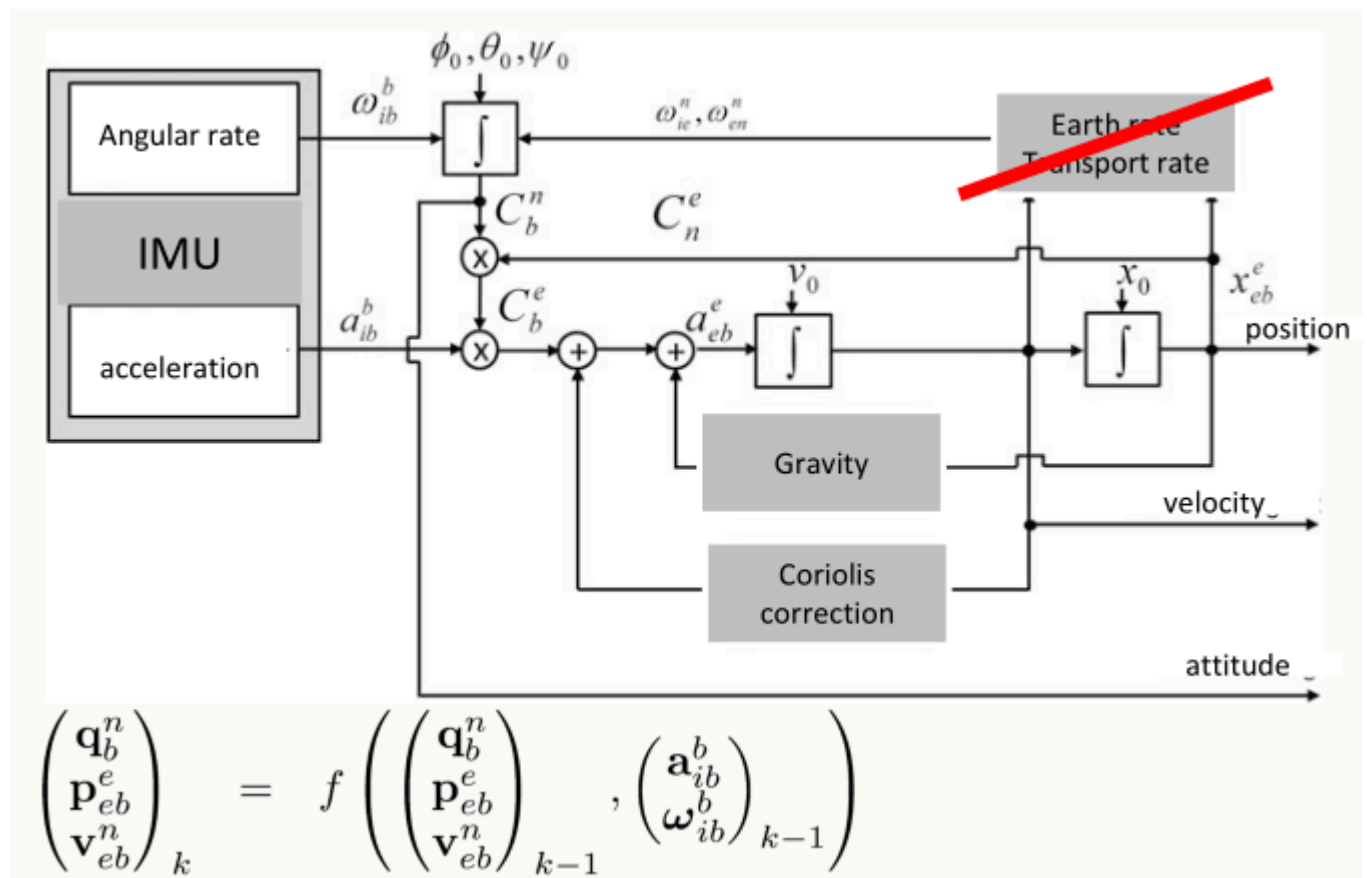
- Combine 1 + 2
- often used

UAV Navigation Sensors

- Standard UAV sensors
- Low-cost IMU for stability control
- Low-cost code-based GNSS (several meters accuracy)
- Sensors used for navigation and mission control
- Can be used for image tagging
 - Time-sync and calibration not critical:
 - Accuracy anyway ~m
- Needs ground control points for precise **indirect georeferencing**
- High-end IMU/GNSS usually used for **direct georeferencing** of laser scans
- Same principle as in mobile laser scanning



Simplified Inertial Navigation



- GNSS: $z_{GPS} = p^n + q_b^n \cdot l^b \cdot q_b^{n,-1}$

- with l^b
- Earth Magnetic Field: $z_{mag} = q_b^{n,-1} \cdot b_{earth}^n \cdot q_b^n$
 - l_{base}^b as local GNSS baseline instead of b_{earth}^n

UAV Application

- Georeferenced Point Cloud
- Observe plant parameters (e.g. Canopy height) during growing season
 - Measure before emergence and at different dates
 - calculate differences
- Building industry, progress documentation
- Surveying and Mapping
 - e.g. Image Based point clouds example